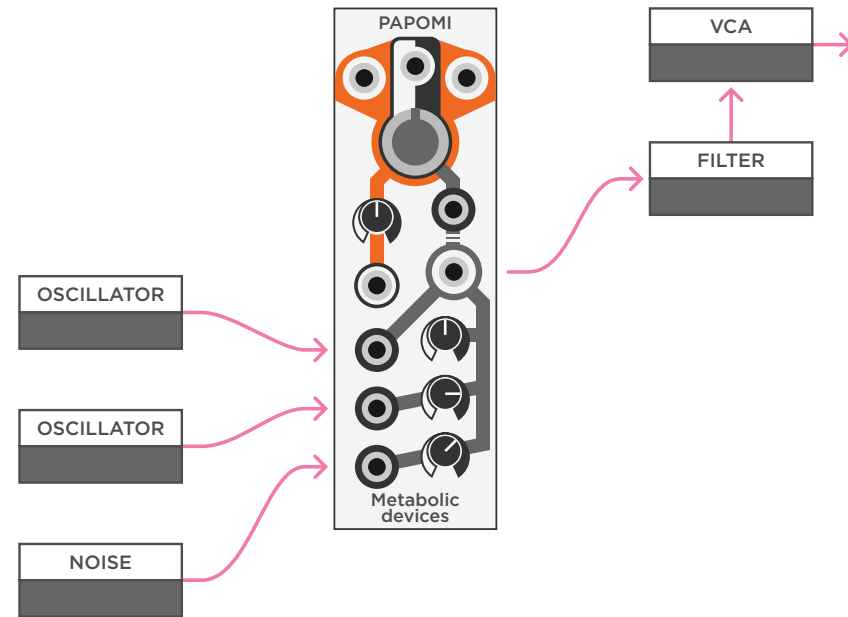


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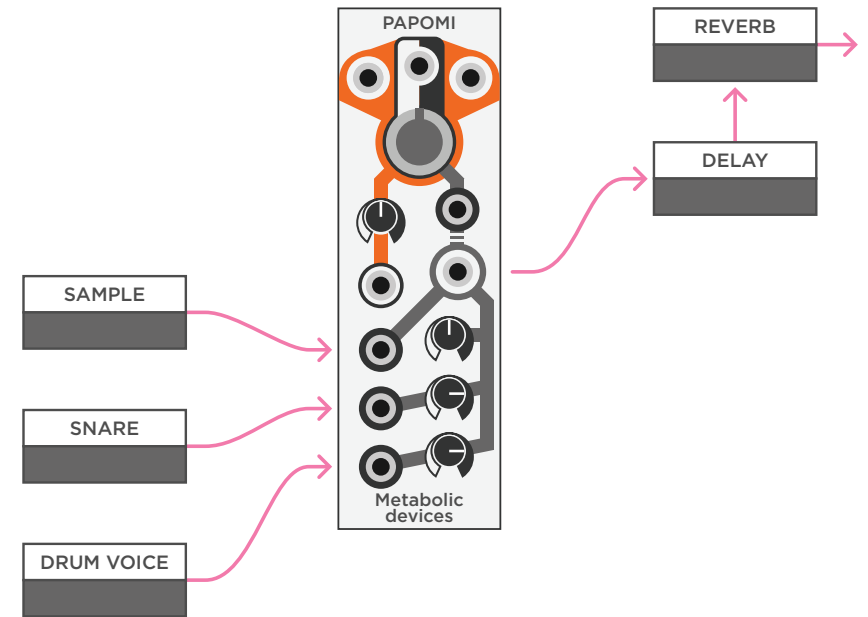
Mixing Audio

Audio
Control Voltage
Trigger / Gate



(a)

a - Here the Papomi is used as an audio mixer, to mix different sources together, like oscillators and noise. Then the mixed audio can be sent to a filter and a VCA, to create a simple mono-synth setup.



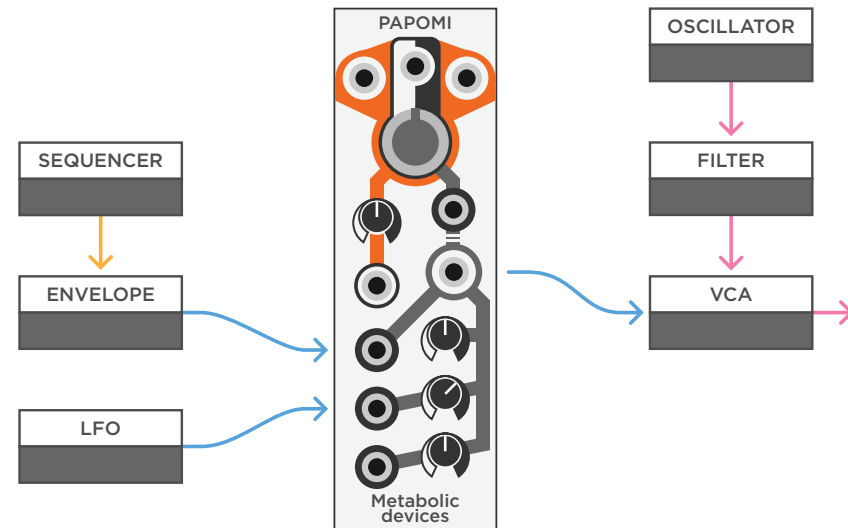
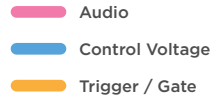
(b)

b - You can also use audio mixers in complex or live patches. For example to mix a few different sound sources like sample players and drum modules together, to create a drum bus. Then you can send the mix through an effects chain.

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Mixing Control Voltages

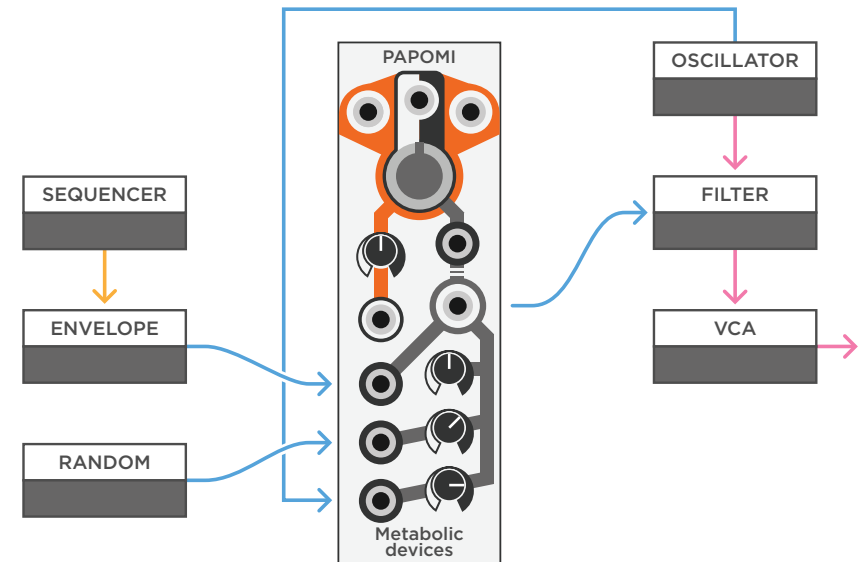


(a)

a - Here the mixer is used to mix control voltages. In this example there is a simple oscillator, filter, and VCA voice. If you want to use a gate with an envelope to open that VCA, but you also want a bit of tremolo created by an LFO, you can use the mixer to mix those signals.

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TECH TALK

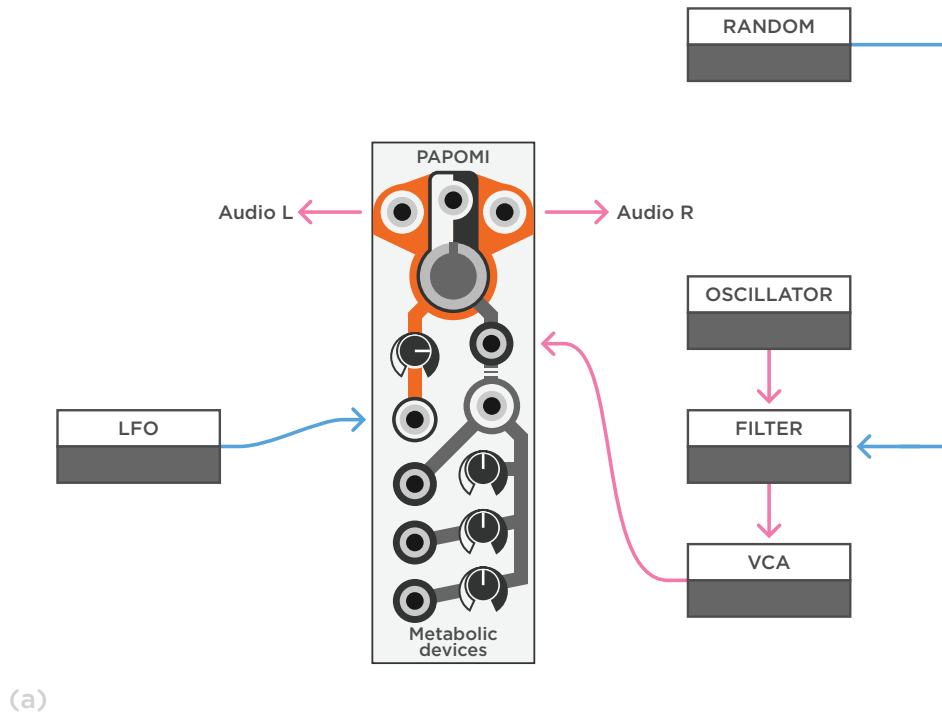


(b)

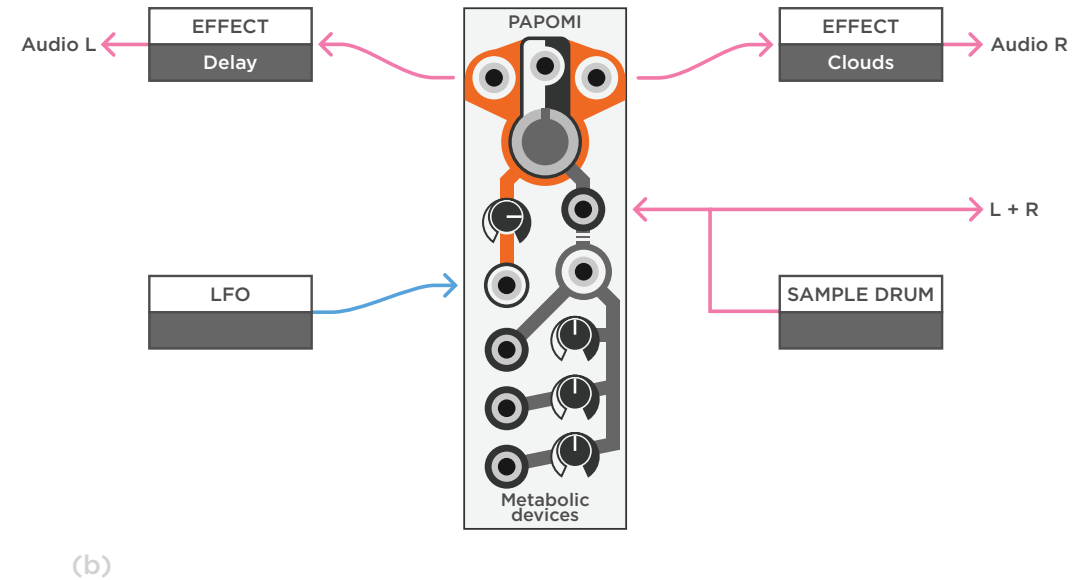
b - In this example there are three signals mixed together to modulate the filter. An envelope, a slow random voltage, and some audio rate modulation from the oscillator in the same voice, which is a great way to add texture.

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



a - You can feed audio from a simple voice with some modulation into the panner input. Then use something like a simple sinewave LFO, to sweep the audio from left to right. You can use the bipolar attenuator to adjust the amount of stereo spread.

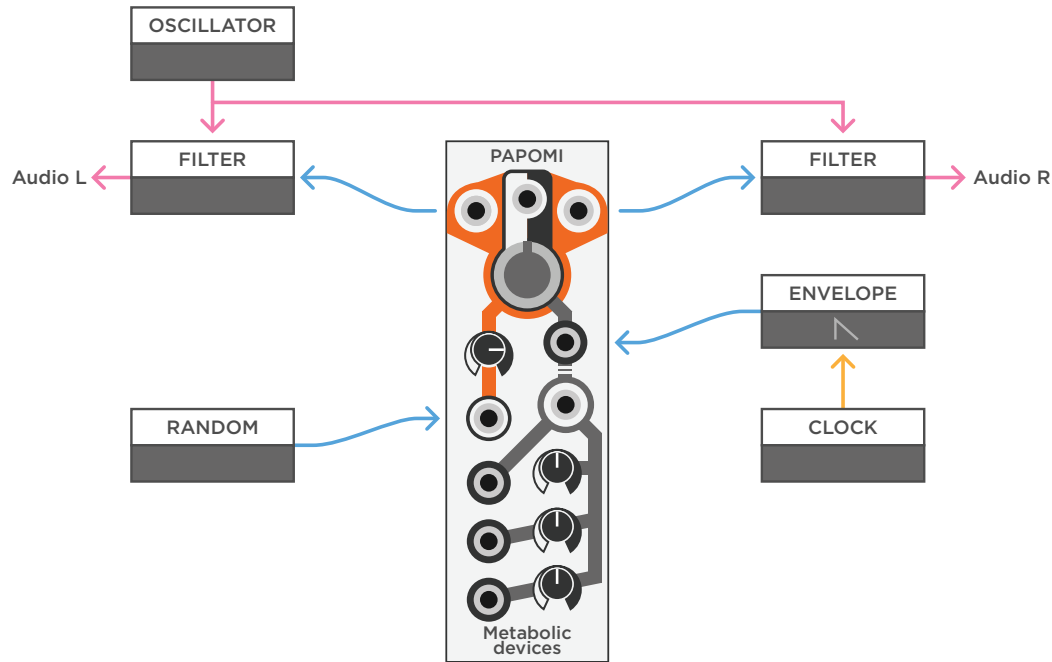


b - You can also use a panner to sweep a sound-source, like a synth voice, or perhaps a nice drum-loop, between two different effects. For example, one with a fast delay and some distortion, and the other with a slower one with some pitch shifting. A slow sinewave LFO is used to pan the signal.

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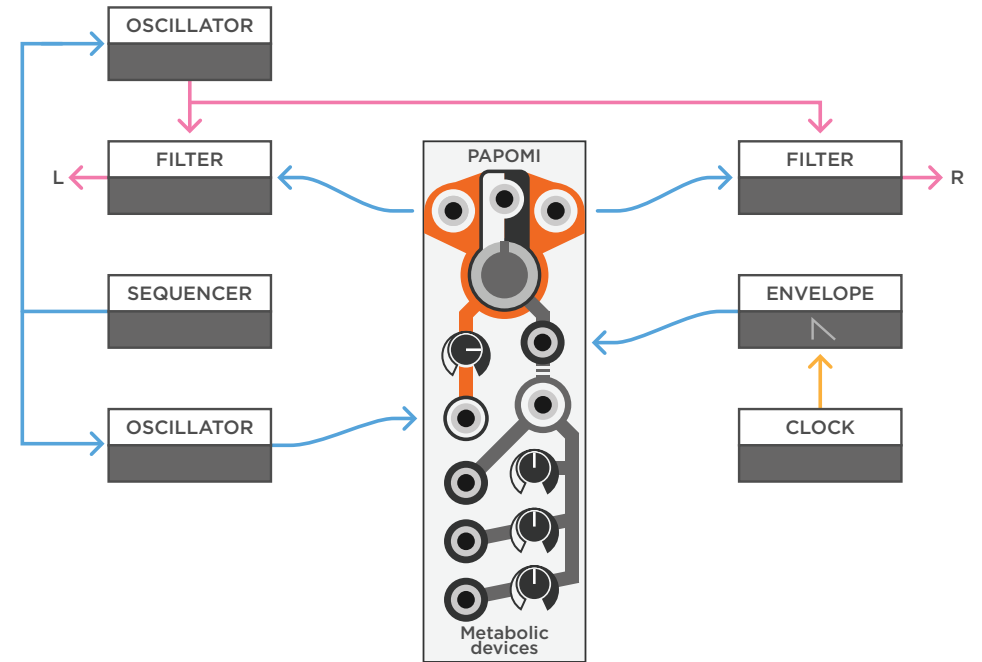
Panning Control Voltages

— Audio
— Control Voltage
— Trigger / Gate



(a)

a - Another way to create interesting stereo patches, is by multiplying one oscillator to two different filters, to create a left and right stereo signal. Then, feed the Papomi a control voltage made with a steady clock and a short attack decay envelope, and have that panned between the filters with a smooth random voltage.



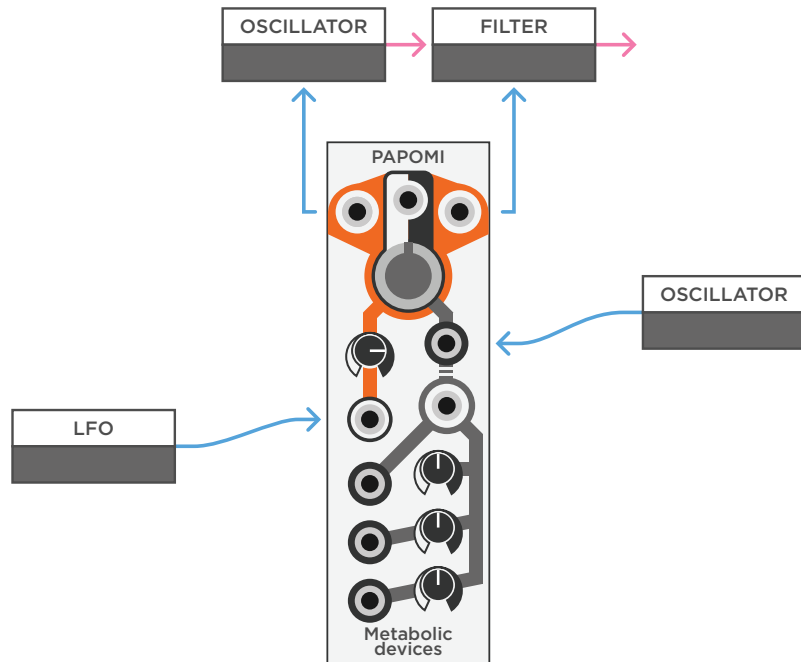
(b)

b - Very often it's interesting to experiment with both slow as well as audio rate modulation. In this patch for example, the smooth random voltage is replaced with another tuned oscillator, and there's a sequencer to control the 1v/oct of both oscillators. This creates a more continuous fat stereo sound.

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Panning Control Voltages

Audio
Control Voltage
Trigger / Gate

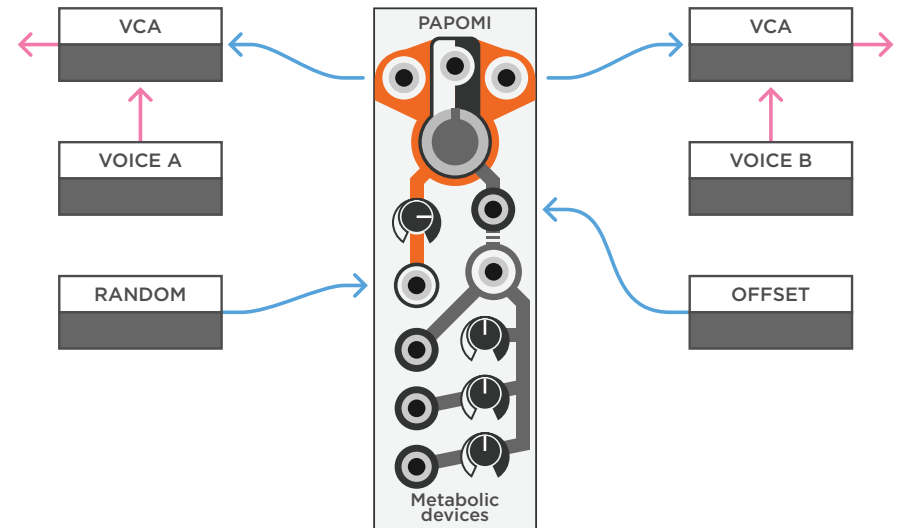


(a)

a - Instead of using a panner to divide an audible sound source, you can achieve interesting effects by just panning modulation signals to different places in your patch. For example, with our simple voice, used as a continuous drone. You can take an audio rate waveform, send it to the panner, and use a modulation source like an envelope, or LFO, to pan that signal between the filter and the oscillator's frequency CV input.

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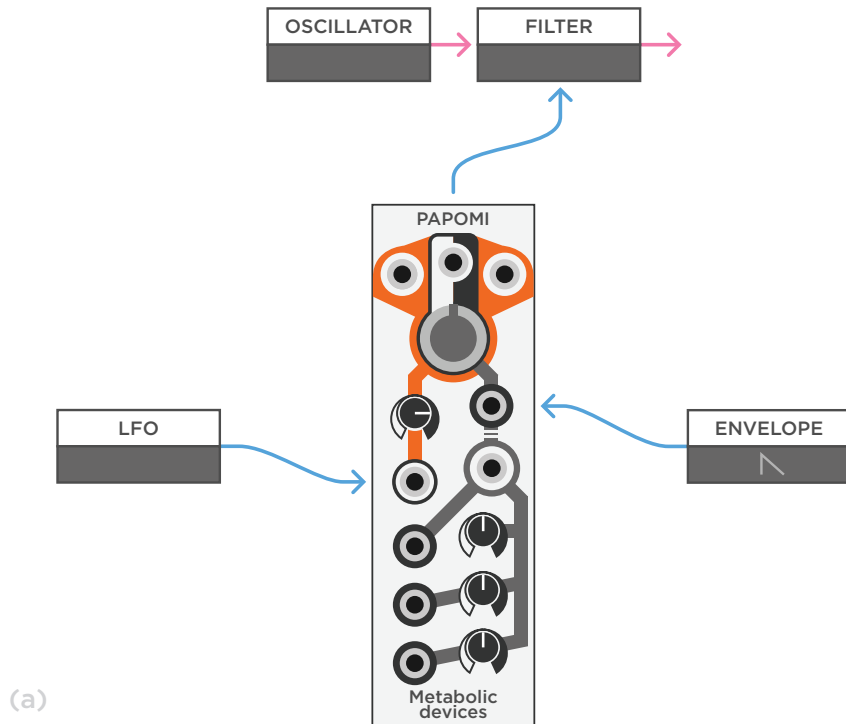
(b)

b - You can feed the panner an offset voltage from the mixer section of the Papomi, use a slow random voltage to pan the signal, and send the outputs to two VCAs to control the volume of two sounds. This way you create a nice organic volume balance between the two.

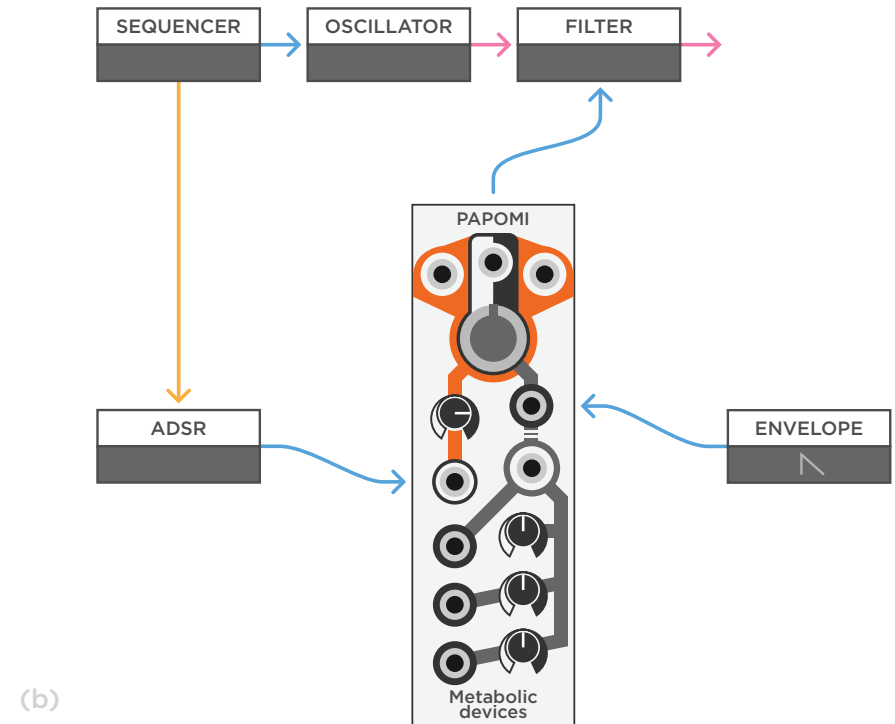
Metabolic Devices Papomi

Polarizing

— Audio
— Control Voltage
— Trigger / Gate



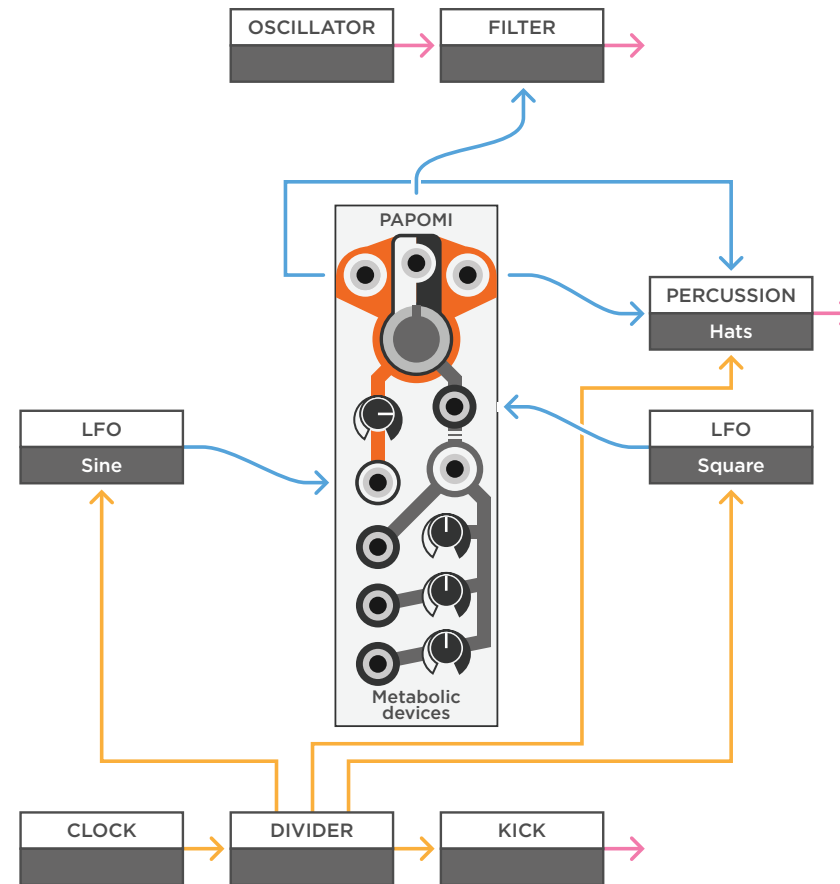
a - Here the polarizer section is used to modulate the filter in a simple voice. If you set the control knob at 12 o'clock and you use a bipolar sinewave LFO to modulate the polarizer, that envelope will sweep between being a positive and negative signal. Or in other words, opening and closing the filter.



b - Especially in this mode, getting the control knob and modulation attenuverter right is important. For example, if you use the same setup, but would like the envelope to have a continuous positive effect, you can turn the control knob clockwise. Then, if you want to use a sequencer with an attack-hold-decay envelope which only sends positive voltages, to sweep the input signal to negative, you can use the bipolar attenuverter on the modulation input to invert the incoming envelope.

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MONOTRAIL TECH TALK



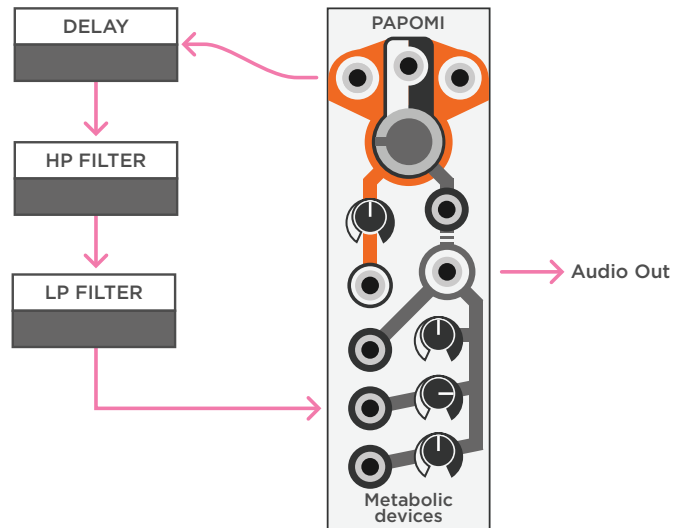
a - In this patch there's a modular clock driving a clock divider. A fast division is used to drive a percussive module, and a slower division to trigger a kick. Then there's a simple constant baseline made with an oscillator and a filter.

Finally, the polarizer output is sent to the filter, so the squarewave LFO creates a shifting accent. And while this patch is going, you can use both the panning outputs to modulate some parameters on the percussive module, which is making some hats.

Metabolic Devices Papomi

Feedback

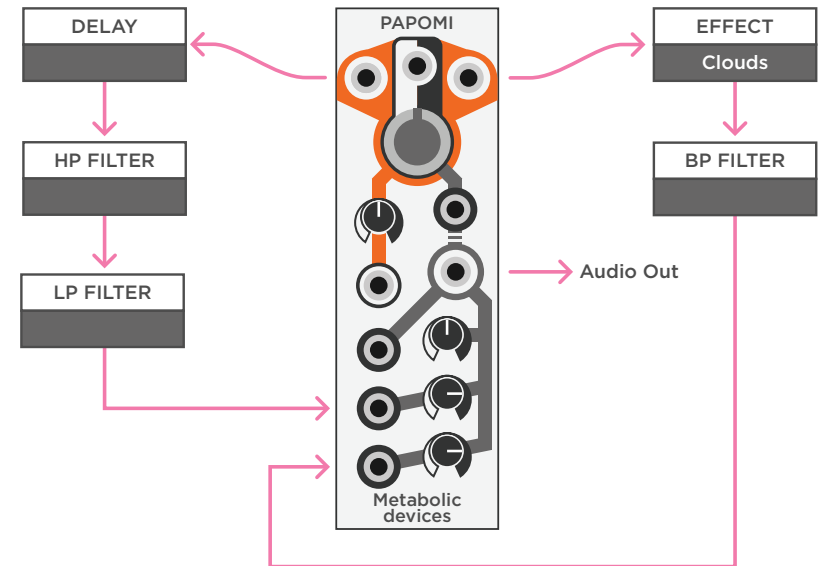
- Audio
- Control Voltage
- Trigger / Gate



(a)

a - Let's set up a simple no input mixing patch. Here there's no audio going into the module. Instead, you can take the left pan output, send it through a delay, and then a high pass and a low pass filter, and then back to one of the inputs. By controlling the main knob, you can send more audio to the left output, and with the input attenuverter, you can control the amount of signal that is sent back to the panner section. This way it's easy to create a feedback loop that goes round and round. However, you can still sculpt the character of the sound because the sound is altered with the delay and filters every times it passes through the loop.

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(b)

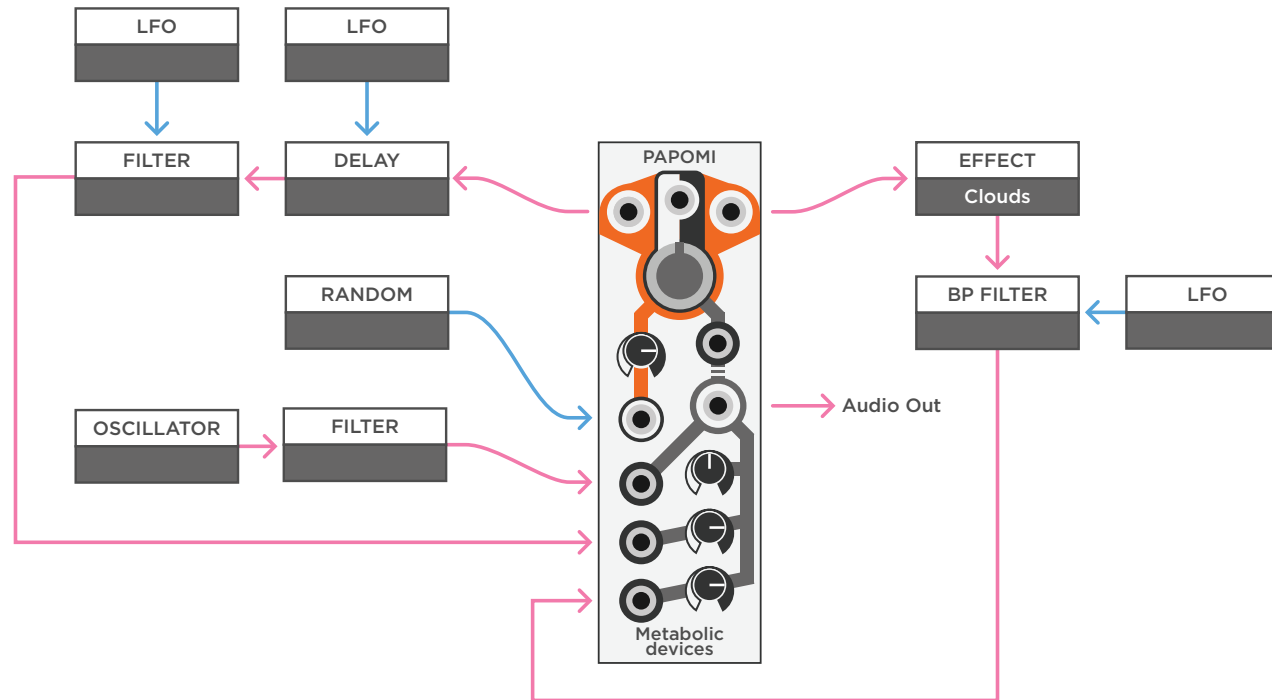
b - You can expand this no input patch by using the right output to create a second feedback loop, for example through the Clouds in pitch shifting mode, and a bandpass filter. Just by controlling the knobs on the Papomi, filters, and effects, you can guide the audio and create different feedback loops by hand.

Metabolic Devices Papomi

Feedback

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - Here you see the same concept, but in a more controlled automated patch. In this case there is a constant drone, made with a single oscillator and filter, going into the unity input. And then the same two feedback loops as used in the last patch, but this time the panner is controlled with a slow random voltage, sweeping the audio between the feedback loops. This way you can create self-playing ambient or drone feedback patches.

You can experiment with different effects to change the sound. And you can expand it as much as you like. In this example a few simple LFOs bring extra movement to the effects and filters in the feedback loops.